Introduction

Programming: Everyday Decision-Making Algorithms

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Today in one sentence

In this course you learn how computer scientists approach everyday decisions, and how to turn these ideas into Python code.

Why it matters

More and more business decisions rely on data and algorithms. You don't need to become a programmer, but understanding how algorithms think, and writing a few lines of code yourself, makes you a better decision maker. No prior knowledge required.

Roadmap

1. About this course: topics, format and how you pass
2. How to learn programming
3. Setting up Python

About this course: topics, format and how you pass

Six topics, one per session

- **I:** Optimal Stopping
- **II:** Explore & Exploit
- **III:** Caching
- **IV:** Scheduling
- **V:** Randomness
- **VI:** Computational Kindness

Take part, it pays off

- **Participate actively**: it makes the course easier and more fun
- Inspired by the book Algorithms to Live By[1]
- All material is online: courses.nilsroemer.com

Each session: lecture, then tutorial

- **Lecture:** we discuss an algorithm for everyday decisions
- **Tutorial:** step-by-step assignments, solved together in groups

[1]Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.

- **Difficulty:** depends on your programming experience

> Tip
>
> No worries, we will help you out if you have any questions!

How you pass

- **Pass/fail course** without exams
- 75% attendance required
- **Two assignments** on algorithmic problems from the course
- **Short final presentation** of your prototype in the last lecture
- You can work in groups of **up to three students**

How to hand in assignments

- Each group submits **one solution** per assignment
- Hand in your **working notebooks**
- Due by the **end of the quarter**, earlier is better
- At least **50%** of the solutions have to be correct

You learn Python in the tutorials

We will mostly **not cover Python during the lectures!**

Question: Does anybody know why?

- The best way to learn is by **doing**
- Lectures focus on the decision-making algorithms
- You learn Python by doing the tutorials

Programming is like a new language

- At first it might be a little bit overwhelming
- First you get used to it and learn words
- Later you apply it and see results
- Important: *Practice, practice, practice!*

What you take home

- The basics of programming
- **Algorithmic thinking**: how computer scientists approach a problem
- Methods you can **apply** to practical problems

Why Python?

- **Simple syntax**: easy to read and to learn
- **Versatile**: web, data analysis, artificial intelligence
- **Big community**: help and documentation everywhere

Learn first, then use AI

- In the tutorials, write the code yourself: that's how you learn
- At the end of the course, you build a small project **with AI support**
- Until then: if you use AI, ask it to **explain**, not to solve

How to learn programming

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Three habits that work

1. Be present: attend the lecture and solve the tutorials
2. Put in some work: repeat code and try to understand it
3. Do coding: run code, play around, modify, and solve

Don't give up!

- Programming is problem solving, errors are normal
- Expect to **stretch** your comfort zone

Setting up Python

Roadmap

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VS Code is where you write code

- Like **Word for text**, but for code: an app to **write, run and find mistakes** in code
- Apps like this are called IDEs (Integrated Development Environments)
- We use VS Code for the **whole course**

The terminal: typing instead of clicking

- A window where you **type a command**, press Enter, and the computer **answers with text**
- No menus, no buttons: the computer does exactly what you typed
- We need it only for the **setup**: a handful of commands, all in the guide to copy

`uv` installs Python for us

- `uv` is a small **helper tool**: it installs Python and everything Python needs
- Everyone gets the **same Python version**, so code works the same on every laptop
- You install `uv` **once**, then it does the rest for you

Notebooks mix text and code

- A notebook is a document with **text cells** and **code cells**
- You run a code cell, and the result appears **right below it**
- All tutorials are notebooks: **read, run, change, try again**

Now we install together

Every step, every command, every click is in the guide:

courses.nilsroemer.com/general/uv.html

Also linked on the course website, under Installation Guide.

- Follow the steps **in order**, don't skip any
- After each step: a small **check**
- We stop at four **checkpoints** and compare

Checkpoint 1: VS Code is open with both extensions

- VS Code shows its **welcome page**
- Under **Extensions**, both **Python** and **Jupyter** say "Installed"

Guide: Steps 1 and 2

Checkpoint 2: The terminal shows a version number for `uv`

- Your folder `programming-course` is open in VS Code
- A terminal is open at the bottom, and it answers to `uv` with a line like `uv 0.11.26`

Guide: Steps 3 to 5

Checkpoint 3: You see Hello from programming-course!

- Python 3.13 is installed
- Your folder is a Python project, and the example program prints `Hello from programming-course!`

Guide: Steps 6 and 7

Checkpoint 4: Your first notebook shows Hello

- The notebook uses the **.venv** kernel with **Python 3.13**
- Below the cell with `print("Hello")`, you see `Hello`

Guide: Steps 8 and 9

Tip

That was the hardest part today! Step 10 of the guide: open a tutorial as a notebook.

What we covered

About this course: topics, format and how you pass

Six topics, each with a lecture and a Python session. You pass with 75% attendance, two assignments and a final presentation.

How to learn programming

Programming is learned by doing: small steps, regular practice, and don't give up.

Setting up Python

Your laptop is ready: VS Code, Python installed with uv, and notebooks. Every step is in the Installation Guide.

Why it matters

More and more business decisions rely on data and algorithms. You don't need to become a programmer, but understanding how algorithms think, and writing a few lines of code yourself, makes you a better decision maker.

Learn by doing.

Any questions

so far?

After the break: Optimal Stopping

- The "Secretary Problem" and the 37% rule
- When to stop searching and make a decision
- Testing the rule ourselves with a small simulation

Note

That's it for our introduction!
Let's have a short break and then continue with our first topic.

Literature

Interesting literature to start

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.[2]

Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Free online
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

> **i** Note
>
> Think Python is free online and a great start. Schrödinger programmiert Python is a playful alternative in German.

More Literature

For more interesting literature, take a look at the literature list of this course.

[2] The main inspiration for this lecture. We have read it and discussed it in depth, always wanting to translate it into a course.